Cogitator Validator: a Pregroup Decision Tree over the `prime_kernel_homology.ipynb` Signal Flow

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Abstract

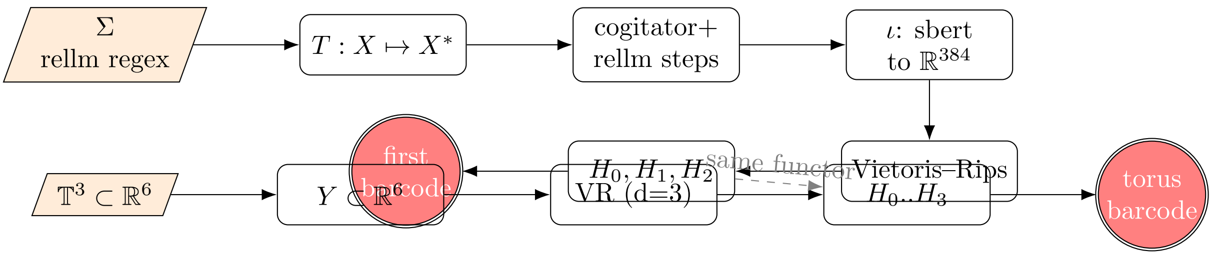
We document a skill plugin that wraps the `prime_kernel_homology.ipynb` pipeline as a reusable Lean checker validator. The skill manages a pregroup-grammar decision tree over eleven kernel checkers (`lean4lean`, `nanoda`, `sonanoda`, `official`, `mini`, `nanobruijn`, `still-nanoda`, `parse-only`, `always-accept`, `always-decline`, `always-reject`) on the test set $\{\text{prime-}p : p \in \{2, 3, 5\}\}$ and on prospective extensions $\{7, 11, 13\}$. The validation core is persistent homology of an `sbert`-embedded point cloud of rellm-tagged cogitator reasoning steps, with the 3-torus $\mathbb{T}^3 \subset \mathbb{R}^6$ acting as the reference topology (Betti pattern $(1, 3, 3, 1)$). A quantum phase estimation (QPE) Betti estimator from `betti_qiskit.py` provides the high-dimensional fallback.

1 Session record

This LaTeX file records the work done in the conversation that produced `cogitator-validator/`. The session deliverables are:

1. A synopsis explaining how the torus barcode glean insights into deductive reasoning steps and into the validity of each separate kernel checker for the prime-number proof test.
2. A philosophical analysis of time estimates for the first five primes ($p \in \{2, 3, 5, 7, 11\}$) under a linear `decide`-step cost model $t_c(p) = a_c + b_c\,(p - 2)$, with explicit refusal to extrapolate beyond $p = 11$.
3. A reading of the vectorisation pipeline as a Vietoris–Rips persistent-homology filtration, with `nx.dodecahedral_graph()` as a symbolic stand-in for the first-loop simplex.
4. A reading of the sublevel filtration as the operation that joins the primary table `df_arena` (kernel verdicts) with the secondary table `df_steps` (parsed rellm messages).
5. A pregroup-grammar decision tree showing how parsed proofs validate the correctness of each (checker, prime) cell.
6. This skill plugin (this folder): plugin manifest, skill markdown, plan, this LaTeX record, and a basic logo.

2 Signal flow



3 Pregroup decision tree

A goal g admits left adjoint g^L (subquestions that precede it) and right adjoint g^R (answers that discharge it). Acceptance is the empty string 1 under the reductions $g^L g \to 1$ and $gg^R \to 1$. The verifier walks:

```
[ROOT] prove(theorem)                                   :: g
 |
 Q1  in-scope?                                          :: g^L
 |   yes -> continue;  no -> reject
 |
 Q2  kernel verdict cached?                             :: g^L
 |   yes -> load matrix M;  no -> run kernel arena
 |
 Q3  parsed_steps.csv has this (checker, test) cell?    :: g^L
 |   yes -> sbert + VR + PH;  no -> run cogitator
 |
 Q4  reasoning Betti table                              :: g^R partial
 |   beta_0 -> #methods; beta_1 -> shared loops;
 |   beta_2 -> three-way chambers
 |
 Q5  does S_rellm_consistency.lean elaborate?           :: g^L
 |   yes -> 1-cat OK;  no -> stop
 |
 Q6  rank checkers                                      :: g^R -> 1
```

4 Relationship with the Betti phase estimator

Let $N = |X|$ be the number of rows in `parsed_steps.csv`. The clique complex $K(\varepsilon) \subset 2^X$ has $|S_k| \le \binom{N}{k+1}$. The combinatorial Laplacian $\Delta_k = \partial_{k+1}\partial_{k+1}^\top + \partial_k^\top \partial_k$ acts on $\mathbb{R}^{|S_k|}$ and the k-th Betti number is $\beta_k = \dim\ker\Delta_k$.

Classical dense eigendecomposition costs $O(|S_k|^3)$. QPE on $U = \exp(2\pi i\,\Delta_k/\lambda)$ produces a phase-0 measurement with probability proportional to $\beta_k/|S_k|$; with $O(|S_k|/\beta_k)$ samples the estimator converges. For sparse Δ_k the per-sample cost is $\mathrm{poly}(\log|S_k|, 1/\varepsilon)$.

Quantum advantage in this scenario

There is no exponential speedup at the current scale of $N = 122$ rows. The genuine advantages of routing through Qiskit are:

- Polynomial speedup on sparse Δ_k at high k, relevant once $|X| > 256$ or once we go to H_3/H_4 of the 1-skeleton.
- No need to materialise Δ_k in RAM; QPE only requires block-encoding access.

- Drop-in compatibility with the 1-categorical signal flow, certified by `S_rellm_consistency.lean`, so the quantum estimator does not change the functor's identity.
- Live hardware path through `QiskitRuntimeService`.

The routing rule embedded in the skill is:

$$\text{use\_qpe} \iff (|X| > 256) \vee (\text{max-dim} \geq 3) \vee (\text{density}(\Delta_k) < 5\%).$$

5 Time-estimate philosophy for the first five primes

We hold ground truth on $p \in \{2, 3, 5\}$ from `lean-kernel-arena`. Three governing forces shape the extrapolation:

- *Kernel checking time* is approximately linear in $p - 2$ (the `Nat.mod` trial-division count). For $p = 7$ expect $\sim 5/3 \cdot t_5$, for $p = 11$ expect $\sim 9/3 \cdot t_5$ per checker.
- *Cogitator wall-time* is roughly constant per (*checker*, *prime*) cell (~ 60 s), so adding two primes doubles total cogitator cost.
- *QPE/TDA wall-time* is decoupled from p; it scales with $|S_k|$, the simplex count, which grows with the added vertices.

Three-point linear fits are exact-fit and offer no statistical confidence; the skill refuses to extrapolate beyond $p = 11$.

6 Files produced

- `.claude-plugin/plugin.json`
- `.claude-plugin/marketplace.json`
- `skills/cogitator-validator/SKILL.md`
- `docs/plan.md`
- `docs/implementation-record.tex` (this file)
- `docs/logo.svg`
- `README.md`